An On-Chain Settlement Layer for Peer-to-Peer Energy Trading under Thailand's Enhanced Single Buyer Model

GridTokenX: Solana/Anchor Programs and Runtime Architecture

Abstract — The growing adoption of renewable energy has yielded participants with surplus generation capacity who currently cannot engage in peer-to-peer (P2P) trading under Thailand's Enhanced Single Buyer model. This article presents an on-chain settlement layer for P2P energy trading in which transaction settlement executes on a Solana cluster. The system features a consortium governance structure mapped to Thailand's grid operators (EGAT, MEA, PEA), an application layer implemented as a set of Anchor smart-contract programs, and consensus and finality driven by Solana's PoH/Tower-BFT engine. The market design incorporates a hybrid continuous double auction (CDA) that matches the on-chain order book in real time for immediate trades, such as BESS reserve dispatch, alongside a uniform-price auction that establishes a single clearing price over 15-minute intervals. We evaluate the smart-contract programs using a transaction-level benchmark on a single-node validator (Apple M2, Agave 3.1.10). Across a TPC-C concurrency sweep (500 transactions, 50% NewOrder / 50% Payment), all transactions confirmed with zero failures. Throughput scaled super-linearly with in-flight concurrency, reaching ≈74 $\text{tx}\cdot\text{s}^{-1}$ at a concurrency of 40 with no saturation knee across the swept range. Compute cost remained flat at ≈23 k $\text{CU}\cdot\text{tx}^{-1}$ across all levels, indicating that the performance ceiling is dictated by single-node block production and write-lock serialisation rather than on-chain execution. That serialisation dominates any path funnelled through a single write-locked account: single-signer mint issuance holds at only ≈5.33 $\text{mint}\cdot\text{s}^{-1}$ regardless of device count, and single-fee-payer settlement at ≈0.6 $\text{tx}\cdot\text{s}^{-1}$. We identify multi-signer fee-payer pooling, together with sharded collector accounts, as the primary lever for issuance- and settlement-side throughput.

Keywords — peer-to-peer energy trading; blockchain settlement; Solana; continuous double auction; uniform-price auction; Thailand Enhanced Single Buyer.

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1. Introduction

Peer-to-peer (P2P) energy trading enables prosumers — households and firms that both produce and consume electricity — to settle surplus generation directly with one another rather than solely through a utility. In Thailand this exchange is currently foreclosed: under the Enhanced Single Buyer (ESB) model all wholesale electricity is procured through a single national off-taker, leaving prosumers with surplus generation no direct route to trade with one another. Realising such an exchange on a public ledger raises three requirements that a general-purpose blockchain does not satisfy by default: (i) **throughput**, because metering telemetry and order flow arrive continuously from many independent devices; (ii) **settlement integrity**, because a trade must be provably authorised by both counterparties yet cannot be allowed to replay; and (iii) **physical backing**, because a token that claims to represent a kilowatt-hour must be tied to an attested, real-world measurement rather than minted at will.

This paper describes GridTokenX, an on-chain energy-trading platform built as a set of Anchor programs on a permissioned Proof-of-Authority (PoA) Solana cluster. The design exploits the Solana Virtual Machine (SVM) execution model directly: it partitions all hot-path state into per-entity program-derived accounts (PDAs) so that unrelated meters, orders, and registrations never contend for the same write lock and therefore execute in parallel under Sealevel. The market runs a hybrid double auction: a continuous double auction (CDA) maintains an on-chain order book and matches compatible orders in real time (`sharded_match_orders`) for immediate trades, such as battery-energy-storage reserve dispatch, while a uniform-price auction clears accumulated bids at a single price over 15-minute epochs coordinated by the oracle's market-clearing trigger. Matches produced by the off-chain engine settle on-chain through native Ed25519 signature verification and per-order replay nullifiers, so the chain never holds custody of the matching engine yet still enforces authorisation and single-use settlement. Token minting is gated behind a registered set of Renewable Energy Certificate (REC) validators, binding issuance to attested generation. An accompanying treasury program provides a Thai-baht-pegged stablecoin (THBG) with reserve-attested peg invariants and a MasterChef-style staking accumulator, giving the market a baht-denominated settlement unit.

Governance follows a permissioned consortium model. The platform authority named in the governance program is intended to be held jointly by Thailand's grid operators — the Electricity Generating Authority (EGAT), the Metropolitan Electricity Authority (MEA), and the Provincial Electricity Authority (PEA) — under a two-step authority-transfer discipline (§6.2), so that protocol administration rests with the same institutions that operate the physical grid, while block production runs on a permissioned Proof-of-Authority validator set (§8).

The contributions of this work are:

1. A Sealevel-parallel account architecture in which every high-frequency write — meter telemetry, order placement, settlement — targets a per-entity PDA, with global aggregates kept deliberately stale and reconciled by periodic admin instructions (§4.2).
2. An off-chain-matched, on-chain-settled trade protocol that verifies buyer and seller Ed25519 signatures through instruction-sysvar introspection and prevents replay with per-order nullifier PDAs (§6.3).
3. A REC-validator gating scheme that ties token minting to attested physical energy (§6.4), together with a two-step authority-transfer mechanism for application-layer PoA governance (§6.2).
4. A treasury design with formally stated peg and collateral invariants and a precise, integer-only economic model for price formation, settlement, fees, wheeling charges, and staking rewards (§11).

The remainder of the paper is organised as follows. Section 2 states the methodology — the design principles, the implementation stack, and the evaluation approach. Section 3 presents the execution stack from off-chain services down to the deployed programs. Section 4 develops the SVM execution model — the account/lock discipline, the compute budget, and zero-copy state. Section 5 describes cross-program invocation between the programs. Section 6 sets out the security and trust model. Section 7 traces an end-to-end market cycle, and Section 8 covers the consensus and validator topology. Section 9 reports the key empirical results and discusses their implications. Section 10 is a reading map to the wider documentation, and Section 11 gives the formal price-formation, settlement, and economic equations, each cited to its implementing source line.

2. Methodology

This work follows a **design-and-measure** method: the platform is constructed as a set of on-chain programs whose structure is dictated by the execution semantics of the target runtime, and every architectural claim is grounded in the implementing source and, where behavioural, in a reproducible measurement. This section states the design principles that constrain the implementation (§2.1), the implementation stack itself (§2.2), and the evaluation harnesses used to verify functional correctness and quantify runtime cost (§2.3).

2.1 Design principles

The design is governed by four principles, each a direct consequence of the SVM account model rather than a stylistic preference.

1. **Execution-model-first partitioning.** Because the runtime serialises any two transactions that write the same account, all high-frequency state is placed in per-entity PDAs, and shared configuration is treated as read-only on hot paths. Parallelism is therefore a property of the data layout, not of an added concurrency mechanism (developed in §4.2).
2. **State in accounts, logic in stateless programs.** Programs hold no mutable state; all protocol state lives in program-owned accounts, which keeps authorisation reducible to the runtime ownership rule.
3. **Defensive integer arithmetic.** All value-bearing computations use checked or saturating integer operations over `u128` intermediates, and every program enables `overflow-checks = true` in its release profile, so an arithmetic overflow aborts the transaction rather than wrapping silently.
4. **Cite, do not assert.** Each architectural statement in this paper is anchored to a specific source location (`path:line`); a claim without such an anchor is treated as a hypothesis rather than a result.

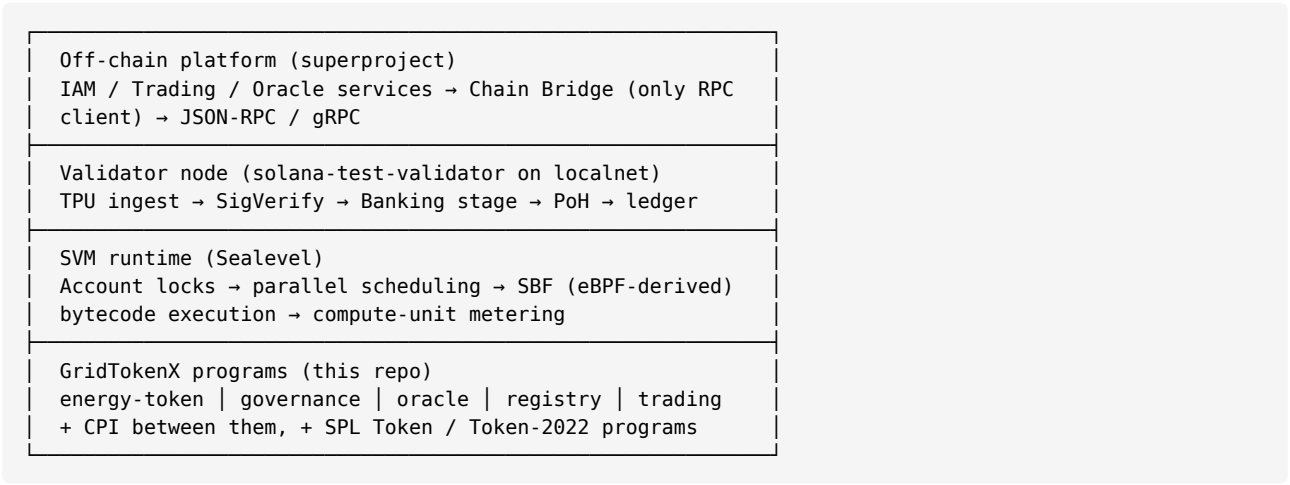
2.2 Implementation

The platform is implemented in Rust and compiled to SBF bytecode with the Anchor framework (`anchor-lang` and `anchor-spl` version 1.0.0). It comprises five core programs — `energy-token`, `governance`, `oracle`, `registry`, and `trading` — a `treasury` program providing the pegged settlement unit, and two benchmark crates (`blockbench` and `tpc-benchmark`); shared functionality is factored into `core` and `compute-debug` crates. Each program is an independent crate rather than a member of a single workspace, and the canonical program identifiers are declared in `Anchor.toml` `[programs.localnet]`. Fungible energy and Renewable Energy Certificate balances are held in Token-2022 mints. State structs are declared zero-copy (`#[account(zero_copy)] #[repr(C)]`) and accessed in place through `AccountLoader`, avoiding full deserialisation of large accounts. Compute-unit consumption is profiled non-invasively by wrapping each handler body in the `compute_fn!` macro (`shared/compute-debug/src/lib.rs:78`), which reports remaining budget on localnet and compiles to a no-op in release builds.

2.3 Evaluation approach

Functional correctness and runtime cost are assessed with a layered set of harnesses, chosen so that most properties can be verified without a validator. In-process suites built on LiteSVM exercise program guards and profile compute-unit cost, with the profiling suites asserting that each instruction remains below the 200,000-CU default budget. Integration suites run under a live validator (`anchor test`, or the standalone runner) to confirm cross-program invocation and settlement end to end, and a Surfpool-based simulation reproduces a mainnet-like environment without a local validator. Throughput and cost under load are characterised with a Criterion micro-benchmark of the matching engine and with the BlockBench and TPC-C workloads; the canonical results are recorded in `BENCHMARKS.md`. A methodological caveat applies to all on-chain measurements: the development topology is a single-node localnet, so reported figures characterise SVM and runtime behaviour (account locking, compute metering, instruction cost) and not consensus throughput, which would require a multi-validator deployment (§8.3).

3. The Execution Stack, Top to Bottom



Key boundary rule (platform-wide, see superproject docs): **no service holds a Solana RPC connection except Chain Bridge**. Everything below the top layer is what this document covers.

4. SVM Execution Model

4.1 Programs are stateless; state lives in accounts

Each program in `programs/*` compiles to SBF bytecode (`anchor build` ⟶ `target/deploy/*.so`) and is deployed at a fixed address (`declare_id!`). A program owns accounts; only the owning program may mutate an account's data. All protocol state — token supply, meter readings, orders, governance config — lives in accounts, not in the programs.

4.2 Transactions declare their account set in advance

A Solana transaction lists every account it will touch and whether each is read-only or writable. The runtime uses this to take **per-account read/write locks** before execution. Two transactions that touch disjoint writable account sets execute **in parallel** on different cores (Sealevel). Two transactions that both write the same account serialise.

This is the most fundamental constraint on the repository's design. Every hot-path write goes to a **per-entity PDA** so that unrelated users/meters/orders never contend:

Hot path	Per-entity account	Seeds	Where
Meter telemetry	`MeterState`	`[b"meter", meter_id]`	`programs/oracle/src/lib.rs:473`
Order placement	`Order`	`[b"order", authority, order_id]`	`programs/trading/src/lib.rs:1435`
Settlement replay guard	`OrderNullifier`	`[b"nullifier", user, order_id]`	`.../settle_offchain.rs:112`
User registration counters	`RegistryShard` ×16	`[b"registry_shard", shard_id]`	`programs/registry/src/lib.rs:770`

Hot path	Per-entity account	Seeds	Where
Settlement escrow	escrow PDA	`[b"escrow", user, currency_mint]`	`.../settle_offchain.rs:140`

Shard selection is deterministic from the signer: `key.to_bytes()[0] % 16` (`programs/registry/src/lib.rs:46`), so a given user always lands on the same shard but the 16 shards absorb concurrent registrations without a global write lock.

The corollary: **global accounts (config, totals) are read-only on hot paths and stale on purpose**. Periodic admin instructions (`aggregate_readings` in oracle, `aggregate_shards` in registry) fold per-entity state back into global totals.

4.3 Compute budget

Every instruction executes under a compute-unit meter (200k CU default per instruction, 1.4M per transaction ceiling). Exceeding it aborts the transaction. This repository profiles CU cost with the `compute_fn!` macro (`shared/compute-debug/src/lib.rs:78`), which logs remaining CU around each handler body on localnet and compiles to a no-op in release. Checkpoints inside long handlers use `compute_checkpoint!` (`shared/compute-debug/src/lib.rs:143`) — e.g. around the registry→energy-token CPI.

4.4 Zero-copy account access

State structs are `#[account(zero_copy)] #[repr(C)]` and accessed through `AccountLoader` (`load()` / `load_mut()` / `load_init()`). Instead of deserialising the whole account into heap objects (Borsh), the program casts the account's byte buffer in place — large accounts (sharded order books, meter state) incur low compute-unit cost. The cost is manual layout discipline: explicit padding fields, no `String` (fixed `[u8; N]` + length byte instead). See invariants 1–2 in ARCHITECTURE.md.

5. Cross-Program Invocation (CPI)

Programs call each other synchronously inside one transaction; the callee inherits the transaction's account locks and compute budget. Two production CPI edges exist:

5.1 registry → energy-token (registration grant)

When a new user registers, the registry program mints the 10 GRX grant by CPI into energy-token. The registry signs the CPI **as its own PDA** using `CpiContext::new_with_signer` with the `[b"registry", bump]` seeds (`programs/registry/src/lib.rs:298`), then calls `energy_token::cpi::mint_tokens_direct` (`programs/registry/src/lib.rs:305`). This is the canonical PDA-signing pattern: no private key exists for the registry PDA; the runtime grants it signer status because the calling program proved the seeds derive to that address.

5.2 trading → governance (PoA config + ERC certificates)

Trading depends on governance with `features = ["cpi"]` and re-exports its types directly: `pub use governance::{ErcCertificate, ErcStatus, GovernanceConfig}` (`programs/trading/src/lib.rs:18`). Settlement validates governance-owned accounts (deserialise + owner check) rather than invoking governance instructions — a read-side coupling, cheaper than a full CPI call.

Both edges are path dependencies in `Cargo.toml`, so Anchor builds callee CPI client code (typed `cpi::` modules) at compile time — no runtime IDL lookup.

6. Security Model

Solana's runtime gives three primitives: **ownership** (only the owning program mutates an account), **signatures** (the runtime verifies tx signers before execution), and **PDAs** (addresses no private key can sign for, derivable only via the owning program). Everything else is program-level policy. This repository adds four mechanisms atop these primitives:

6.1 Anchor constraint checks (account validation)

Account structs declare constraints that Anchor verifies before the handler runs: `Signer<'info>` for required signers, `has_one = authority` for stored-authority matching, `seeds = [...] + bump` for PDA address verification. Governance gates every privileged instruction this way — e.g. `has_one = authority @ GovernanceError::UnauthorizedAuthority` on `IssueErc` (`programs/governance/src/contexts.rs:31`), `ValidateErc` (`programs/governance/src/contexts.rs:90`), `RevokeErc` (`programs/governance/src/contexts.rs:108`), and `UpdateGovernanceConfig` (`programs/governance/src/contexts.rs:149`).

6.2 Application-layer PoA with 2-step authority transfer

The governance program holds a `GovernanceConfig` account naming the platform authority. Authority rotation is two-phase to tolerate accidental key-entry errors:

1. `propose_authority_change` (`programs/governance/src/handlers/authority.rs:13`) — current authority writes `pending_authority` (`authority.rs:34`), guarded against an already-pending proposal (`authority.rs:22`).
2. `approve_authority_change` (`programs/governance/src/handlers/authority.rs:53`) — the **new** key must sign to accept, and only then does `poa_config.authority` flip (`authority.rs:78`).

An incorrectly entered new authority therefore cannot render the protocol inoperable — the incorrect key never signs the approval.

6.3 Off-chain-signed settlement: Ed25519 instruction introspection

The CDA matching engine runs off-chain (Trading Service). Matches settle on-chain via `settle_offchain.rs` without the buyer/seller being transaction signers. Instead:

- The off-chain engine collects **Ed25519 signatures from buyer and seller over the order payload** and places them in the transaction as instructions to Solana's native Ed25519 verification program (which the validator executes — and rejects the transaction if invalid — before the trading instruction runs).
- The trading handler then **introspects the transaction** via the instructions sysvar: `verify_ed25519_signature` (`.../settle_offchain.rs:651`) calls `load_instruction_at_checked` (`settle_offchain.rs:657`), asserts the instruction targets the Ed25519 program (`settle_offchain.rs:660`), and asserts the pubkey embedded in the verified instruction matches the expected order owner (`settle_offchain.rs:669`). Buyer and seller are checked at instruction indexes 0 and 1 (`settle_offchain.rs:310`, `settle_offchain.rs:314`).

This is the standard Solana pattern for "verify an arbitrary off-chain signature on-chain" — the expensive curve math runs in the native program; the application program only proves the verification happened in the same transaction, against the right key and message.

Replay protection comes from `OrderNullifier` PDAs seeded per `(user, order_id)` (`settle_offchain.rs:112`). Each nullifier tracks `filled_amount`; a settlement may only consume the unfilled remainder (`settle_offchain.rs:344`) and increments the fill saturatingly (`settle_offchain.rs:428`), with the nullifier's stored authority re-checked against the payload (`settle_offchain.rs:539`). The same signed order can therefore be partially filled across transactions but never over-filled or replayed.

6.4 REC-validator gating on mint

GRID tokens represent physical energy (1 kWh = 1 GRID), so minting is gated behind registered REC validators in energy-token. Validators are added/removed by the admin (`programs/energy-token/src/lib.rs:280`, `lib.rs:313`); when any validator is registered (`rec_validators_count > 0`, `lib.rs:119`), mint paths require the signing key to appear in the validator set (`lib.rs:127–128`), with the same gate on the generation-mint path (`lib.rs:203`).

6.5 Trust boundary summary

Boundary	Enforced by
Who may mutate an account	Runtime ownership rule (program-owned accounts)
Who may invoke privileged instructions	Anchor `Signer` + `has_one` against `GovernanceConfig` / stored authorities
Whether a trade was authorised	Ed25519 native-program verification + sysvar introspection
Whether a trade can be replayed	`OrderNullifier` PDA per (user, order)
Whether energy backing is real	REC validator set in energy-token
Who may even reach the RPC port	Off-chain: Chain Bridge mTLS + RBAC (superproject concern)

7. Protocol Flow (end-to-end runtime view)

A full market cycle touches all five programs:

```
1. Registration   registry: create user PDA on shard (key[0] % 16)
                  └─CPI→ energy-token: mint 10 GRX grant (PDA-signed)
2. Telemetry      oracle: AMI gateway submits readings → per-meter MeterState PDA
                  (parallel across meters; 15-min market-clearing epochs)
```

```
3. Order entry    trading: submit_order → per-order PDA on order shard
4. Matching       OFF-CHAIN: Trading Service CDA engine matches buy/sell,
                  collects buyer+seller Ed25519 signatures
5. Settlement     trading: settle_offchain_match — ed25519 ix at index 0/1,
                  introspection check, nullifier fill update, escrow transfer
                  (reads governance GovernanceConfig / ERC certificates)
6. Token movement energy-token / SPL: GRID + GRX transfers, REC-gated mint/settle
7. Reconciliation admin: aggregate_readings / aggregate_shards fold shard + meter
                  state into global totals (deliberately stale between runs)
```

Steps 2 and 3 are the throughput-critical paths and are exactly the ones designed for Sealevel parallelism (§4.2). Step 5 is the security-critical path (§6.3).

8. Consensus & Validator Topology

8.1 What Solana consensus provides

On any Solana cluster, ordering and finality come from **Proof of History** (a verifiable delay function giving each entry a position in time) feeding **Tower BFT** (a PoH-timestamped variant of practical BFT voting in which validators commit to forks with exponentially growing lockouts). Block production rotates across a leader schedule weighted by stake. The programs in this repository are consensus-agnostic: they see only the SVM account model and cannot tell which consensus produced the block.

8.2 What "permissioned PoA" means here

GridTokenX targets a **permissioned cluster**: the validator set is a closed list of known operators (utility / market-operator nodes) rather than open stake-weighted entry. Solana does not have a separate "PoA mode" — permissioning is operational (who is allowed to run a validator and receive stake delegation), and the **application-layer** PoA lives in the governance program's `GovernanceConfig` (§6.2), which gates protocol administration regardless of who validates blocks. The two layers must be kept distinct:

Layer	Authority	Mechanism
Block production / finality	Permissioned validator operators	PoH + Tower BFT among allowlisted nodes
Protocol administration	`GovernanceConfig.authority`	Governance program checks (§6.1–6.2)
Energy attestation	REC validator set	energy-token gating (§6.4)

8.3 Development topologies

Mode	Nodes	Consensus reality	How
`anchor test` / localnet	1 × `solana-test-validator`	None — single node self-produces blocks; PoH runs but no voting quorum	`anchor test`, or superproject `just solana-up`
Surfpool simnet	0 local validators	Simulated against mainnet state	`npm run simnet` / `npm run simnet:ci`
Target deployment	N permissioned validators	Real Tower BFT among allowlisted operators	Out of scope for this repository

The single-node localnet means dev/test never exercises fork choice, leader rotation, or vote lockouts — only the SVM semantics (§4) are faithfully reproduced. Performance numbers from `BENCHMARKS.md` are therefore SVM/runtime measurements, not consensus-throughput measurements.

9. Key Results and Discussion

This section reports the principal empirical findings and interprets them against the design goals of §2.1. All figures are drawn from the canonical benchmark report `BENCHMARKS.md`. The compute-unit (CU) values are machine-independent and are reproduced from the in-process LiteSVM profiles, whereas the throughput figures are qualified by the single-node caveat of §2.3.

Table 1 summarises the headline findings; each is developed and cited in the subsections that follow. Two throughput limits are reported separately and must not be conflated: the matching/OLTP path scales super-linearly to the swept

ceiling, whereas any path that funnels through one write-locked account (mint issuance, single-fee-payer settlement, single-gateway meter ingest) is bounded by that serialisation, not by execution.

Metric	Measured value	Bottleneck source
Peak throughput (TPC-C proxy, c=40)	≈74.25 tx·s^{-1}	block-time + write-lock on shared accounts
Scaling (c=5⟶c=40)	super-linear (9.4× over 8×), no saturation knee	— (execution is load-independent)
Per-tx compute (all loads)	flat ≈22–24 k CU	load-independent (not execution-bound)
Single-signer mint issuance	≈5.33 mint·s^{-1}	shared `mint` write-lock (device-count independent)
Batch-settle (single fee-payer)	≈0.6 tx·s^{-1}	global collectors + `treasury_state` accumulator write-lock
Meter ingest, 100,000 meters	0.35% delivery loss, ≈13.5 k CU flat	single gateway payer write-lock (not per-meter)
Transaction failure rate (TPC-C sweep)	0% at all concurrency levels	—
Settlement compute cost	121,813 CU·match^{-1} (≈61% budget)	Ed25519 verify + dual-mint escrow

Table 1: Headline empirical results. All values are drawn from the canonical benchmark report `BENCHMARKS.md` (`3cb3388`, Solana 3.1.10, Apple M2, 2026-07-06). CU figures are machine-independent; absolute throughput is qualified by the single-node caveat (§2.3, §8.3).

9.1 Compute-unit cost of on-chain operations

Table 2 summarises the measured CU cost of representative instructions and their share of the 200,000-CU default per-instruction budget. Two facts stand out. First, every control, telemetry, and settlement-recording path costs no more than approximately sixteen thousand CU — at most about eight per cent of the budget — so the high-frequency operations retain ample headroom. Second, the off-chain-match settlement instruction, `settle_offchain_match`, is by a wide margin the most expensive operation at roughly 122,000 CU (about 61% of the budget); its cost is dominated by the native Ed25519 signature verification and instruction-sysvar introspection that authorise the trade (§6.3), not by the settlement arithmetic itself.

Instruction	CU	% of 200k
`do_nothing` (dispatch + signature-verify floor)	769	0.4%
`treasury.record_settlement`	3,301	1.7%
`governance.approve_authority_change`	5,784	2.9%
`treasury.record_settlement_sharded`	5,374	2.7%
`oracle.trigger_market_clearing`	8,390	4.2%
`oracle.submit_meter_reading` (steady-state)	13,376	6.7%
`oracle.submit_meter_reading` (first, inits PDA)	16,050	8.0%
`trading.settle_offchain_match`	122,000	61.0%

Table 2: Measured compute-unit cost of representative instructions and their share of the 200,000-CU default budget. Source: `BENCHMARKS.md` §1, §4, §5, §5b.

The hot telemetry write, `submit_meter_reading`, costs about 16,050 CU on the first submission for a meter — which initialises the meter PDA — and about 13,376 CU in steady state, confirming that the dominant recurring on-chain write sits comfortably inside budget. The governance and oracle control instructions all fall between roughly 5,800 and 11,100 CU.

9.2 Parallelism and throughput under load

Under the TPC-C concurrency sweep, aggregate throughput rises from 7.87 transactions per second (TPS) at concurrency 5 to 74.25 TPS at concurrency 40 — a 9.4-fold increase over an 8-fold rise in concurrency — climbing monotonically

with no saturation knee within the swept range. Crucially, the per-transaction CU cost remains flat at approximately 22,000–24,000 CU across all concurrency levels, which shows that on-chain execution cost is independent of load. The throughput ceiling is therefore imposed by consensus block time and by write-lock serialisation on the few genuinely shared accounts, not by program execution.

The single-node development topology bounds absolute write throughput at approximately 2.0 TPS for any operation that requires a block, because latency is dominated by the 400–600 ms block time rather than by compute. By contrast, a read served directly by the RPC node returns in under a millisecond (≈1,454 TPS), three orders of magnitude faster, as it involves no consensus round-trip. These absolute rates are properties of the single-node harness, not of the protocol, and must not be read as consensus-throughput results (§2.3, §8.3).

A distinct serialisation limit governs token issuance. Settlement mints GRID against a matched trade, and because every settle write-locks the same treasury accumulator and the shared fee, wheeling, and loss collector accounts, issuance does not parallelise across the contributing devices: an open-loop sweep in which a single fee-payer submits settlements holds at approximately 0.6 mint·s^{-1} irrespective of in-flight concurrency and of whether the offered load originates from one device or many. The bottleneck is the shared accumulator, not the signer or the device population — the direct-mint path, by contrast, packs many mints into a single slot from one signer, confirming that raw issuance is limited by the confirmation round-trip rather than by on-chain contention. Two complementary remedies restore parallelism: sharding the collector and accumulator accounts so that concurrent settlements touch disjoint writable state (§9.4), and pooling multiple fee-payers so that submission is not funnelled through a single account. In a Tier-A prototype combining both, per-slot settlement packing rises to approximately 7.5 mint·s^{-1}. We therefore identify multi-signer fee-payer pooling, together with per-shard collector accounts, as the primary lever for settlement-side throughput.

Table 3 consolidates every throughput figure measured in this evaluation. All values are **client-observed confirmed goodput** — successfully confirmed transactions per wall-clock second, measured from burst start to last observed confirmation at `confirmed` commitment, so each figure includes the full pipeline (client signing, RPC ingest, banking-stage execution, block production, and confirmation visibility). None is a consensus- or network-throughput result (§2.3, §8.3); rows measured through different client harnesses (the OLTP proxy's per-transaction confirmation versus the ingest benchmark's raw submission) are not directly comparable with one another.

Path	TPS	Bottleneck / regime
RPC read (no consensus round-trip)	≈1,454	none — RPC node local
Meter ingest, steady (5k–50k meters, sustained)	393–490	single gateway fee-payer write-lock
Meter ingest, steady (100k meters)	322	single gateway fee-payer write-lock
CDA order entry, sharded (10k orders)	180	shared `zone_market` write-lock + fee-payer
TPC-C OLTP proxy (concurrency 5 ⟶ 40)	7.87 ⟶ 74.25	block time + shared-account locks
Sequential single-operation write	≈2.0	400–600 ms block time per round-trip
Settlement mint, Tier-A prototype (sharded + pooled payers)	≈7.5	per-slot packing
Settlement mint, single fee-payer	≈0.6	global collector/accumulator write-lock

Table 3: Consolidated measured throughput. All values are client-observed confirmed goodput on the single-node validator; harnesses differ per row and rows are not mutually comparable. Source: BENCHMARKS.md §3, §9 and the settlement sweeps of §9.2.

The spread spans three orders of magnitude and sorts cleanly by how much shared, write-locked state each path touches: reads lock nothing; meter ingest locks only the gateway payer; order entry additionally write-locks the shared zone-market account; the OLTP proxy locks a few market accounts; and naive settlement locks global accumulators. Order entry makes the ordering principle explicit: it is the cheapest hot-path write measured (≈9.9 k CU, below the ≈13.5 k CU meter write) yet delivers 2.6× lower throughput than meter ingest at the same fleet size, because its account contexts still declare the shared zone-market account writable although the handler mutates only the per-shard account. Throughput on this platform is a function of lock footprint, not of instruction compute cost.

9.3 Meter-telemetry ingest scaling to 100,000 meters

The concurrency sweep above uses a generic-OLTP proxy. To characterise the **actual** telemetry hot path at fleet scale, we constructed a fleet-ingest benchmark (`scripts/bench-meter-throughput.ts`) that emulates an AMI gateway submitting one reading per meter for fleets of 80 up to 100,000 distinct meters against a live single-node validator.

Test design. Each simulated meter has a unique identifier, so each submission targets its own `MeterState` PDA (`[b"meter", meter_id]`) and the write set of any two submissions is disjoint (Sealevel-parallelisable, §4.2). All submissions are signed by one gateway key — the registered `chain_bridge` authority that the oracle program requires (§7 step 2) — which mirrors the production topology in which a single Aggregator Bridge fronts the meter fleet, and which deliberately preserves the shared-fee-payer serialisation the paper identifies as the ingest ceiling. Every run submits **two epochs** of readings: the on-chain rate-limit guard (`min_reading_interval` = 60 s) and the strictly-increasing-timestamp guard force epochs at least 61 s apart, and the split separates the one-off account-creation cost (epoch 1 initialises every meter PDA and pays its rent) from the recurring steady-state write (epoch 2), which is the figure that matters for a fleet in operation. Reading values follow a fixed synthetic pattern; each transaction carries exactly one `submit_meter_reading` instruction.

Measurement. Throughput is client-observed confirmed goodput as in Table 3; latency is send-to-confirmed per transaction; compute units are read post-hoc from transaction metadata of a sample of confirmed transactions. Every non-confirmed transaction is attributed to one of three failure classes: **send-rejected** (the RPC node refused the submission — never entered the validator, no fee), **on-chain error** (executed and failed a program guard — fee paid, failure recorded on the ledger), and **expired** (accepted but never confirmed within 90 s — dropped from the ingest queue or its blockhash aged out; no fee, no trace). The classes have different operational meaning: on-chain errors are the oracle's input validation doing its job, while send-rejected and expired transactions are pure delivery loss, invisible on-chain and recoverable by client retry — the rate-limit and monotonic-timestamp guards make resubmission idempotent: no reading can be double-counted. All fleet sizes use one transport: transactions are pre-signed locally against a cached blockhash, submitted raw without preflight or retry, and confirmed by bulk signature-status polling (1.5 s sweep) under a 3,000-transaction in-flight window, so latency is poll-quantised (±1.5 s). Reading timestamps are taken from the on-chain clock rather than the client's wall clock, matching the clock against which the oracle's freshness guard is evaluated. The formal metric definitions and the full harness parameterisation are given in §11.9. Table 4 reports the result.

Meters	Confirmed / total	Validation- rejected	Delivery loss	Steady CU min	TPS (steady)
80	160 / 160	0%	0%	13,560	47
1,000	1,988 / 2,000	0.40%	0.20%	13,634	267
5,000	9,906 / 10,000	0.48%	0.46%	13,634	393
10,000	19,862 / 20,000	0.46%	0.23%	13,337	462
50,000	99,247 / 100,000	0.47%	0.28%	13,337	490
100,000	198,347 / 200,000	0.48%	0.35%	13,634	322

Table 4: Oracle meter-telemetry ingest scaled from 80 to 100,000 meters on a live single-node validator, with every non-confirmed transaction attributed. Validation-rejected = the oracle's anomaly gate firing on the synthetic value pattern (§11.9); delivery loss = send-rejected + expired, with no client retry. Steady-state CU is the recurring per-meter write; the base path (`CU min`) is the value to cite. Source: `BENCHMARKS.md` §9.

Three results stand out. First, the steady-state base write cost holds at 13.3–13.6 thousand CU across a 1,250-fold increase in meter count, matching the 13,376 CU litesvm figure of §9.1: the per-meter write is $O(1)$ in fleet size, exactly as the per-entity-PDA partitioning predicts, with the residual ±150 CU attributable to meter-identifier byte length rather than fleet size. Second, the loss decomposition separates two very different phenomena. The validation-rejected column is constant at ≈0.47% because the synthetic value pattern deterministically drives that fraction of meters past the anomaly gate (Equation 17) — the same meters are rejected in every epoch, each rejection is fee-paid and recorded, and the bucket demonstrates the oracle's input validation firing correctly under 100,000-meter load. True delivery loss — submissions that vanished in transit — stays at or below 0.46% at every fleet size (0.35% at 200,000 transactions) with no client retry; because resubmission is idempotent, a single retry round would reduce the effective rate to the order of 10^{-5}. Third, throughput does not grow with meter count: small fleets are ramp-limited (an 80-transaction burst never fills the pipeline), sustained rates reach roughly 390–490 TPS between 5,000 and 50,000 meters, and the fleet doubling from 50,000 to 100,000 does not raise it, because every submission is signed by the single gateway fee-payer, which is write-locked on each transaction — as with settlement in §9.2, the ceiling is a shared-account serialisation limit, not per-meter contention. The flat CU confirms the on-chain write itself is already scale-free; converting the disjoint per-meter PDAs into proportional throughput requires the same remedy identified for settlement, namely pooling multiple gateway fee-payers so ingress is not funnelled through one signer.

9.4 Discussion

The measurements support the central design claim of §2.1: partitioning high-frequency state into per-entity PDAs removes execution-side contention, so the residual scaling limit is consensus and lock serialisation rather than compute. The flat CU-versus-concurrency profile is the direct evidence — had hot paths shared a writable account, compute cost or failure rate would have risen with load; neither does. The sharded settlement-recording path reinforces this: `record_settlement_sharded` writes a per-shard PDA at about 5,374 CU instead of contending on a single global counter, converting a would-be serialisation point into an embarrassingly parallel one, at the cost of a periodic off-hot-path aggregation.

Settlement is the natural cost centre, and the 122,000-CU figure quantifies the price of trustless authorisation: verifying two counterparty signatures on-chain and proving that verification through sysvar introspection. This remains a single transaction well within the per-transaction budget, but it explains why batch settlement processes one match per transaction — the Ed25519 signature data for each match cannot be compressed through an address-lookup table — so batching amortises account-setup overhead rather than signature cost. Reducing this figure, for example through succinct proof aggregation over many matches, is the clearest avenue for further compute savings.

The principal limitation of this evaluation is topological. Because all on-chain measurements are taken on a single self-producing validator, they faithfully characterise SVM semantics — account locking, compute metering, and per-instruction cost — but say nothing about consensus throughput, fork choice, or leader rotation, which would require a multi-validator permissioned deployment (§8.3). Establishing sustained end-to-end throughput on such a deployment, and locating the true saturation point above concurrency 40, are left to future work.

10. Reading Map

Question	Read
What are the programs / IDs / invariants?	`ARCHITECTURE.md`
Zero-copy layout, sharding, CU profiling detail	`SKILL.md` (versions/IDs stale — trust `Anchor.toml`)
Benchmark methodology + results	`BENCHMARKS.md`
Settlement math / clearing	`programs/trading/src/instructions/settle_offchain.rs`
Platform-wide rules (Chain Bridge, gateways)	superproject `../CLAUDE.md`, `../ARCHITECTURE.md`

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11. Formal Model: Pricing, Settlement & Economic Equations

All on-chain money math uses integer `checked_mul` with `u128` intermediates — overflow **rejects** the transaction, never clamps. $\lfloor \cdot \rfloor$ denotes the integer floor division the SBF program performs.

11.1 Notation

Symbol	Meaning	Scale
q	matched energy quantity (`match_amount`)	9-dec, 1 kWh $= 10^9$
p	clearing price (`match_price`)	6-dec THBG/kWh
V	trade gross value	6-dec THBG
φ_m	market fee (`market_fee_bps`)	bps
w	wheeling rate (`wheeling_rate_per_kwh`)	6-dec THBG/kWh
ℓ	loss rate (`loss_bps`)	bps
r	peg rate (`grx_per_thbg_rate`)	THBG-minor / whole GRX
φ_s	swap fee (`swap_fee_bps`)	bps
A	staking accumulator (`acc_reward_per_share`)	$\times 10^{12}$
T	total staked GRX (`total_staked`)	atoms

11.2 Price formation

Prices are limit prices quoted in THBG per kilowatt-hour (6-decimal fixed point). The market operates two coordinated mechanisms — a continuous double auction for immediate execution and a uniform-price auction over 15-minute epochs — and both resolve to the same settlement arithmetic of §11.3.

Order admission. Every order carries a strictly positive limit price and must fall within the market's configured band (`programs/trading/src/lib.rs:221`, `lib.rs:226–233`); orders outside $[p_{\min}, p_{\max}]$ are rejected at entry, which bounds the price domain before any matching occurs:

$$p_{\min} \le p \le p_{\max} \tag{1}$$

Continuous double auction (on-chain path). Two orders may match only when they cross (`programs/trading/src/lib.rs:454`):

$$p_b \ge p_s \tag{2}$$

and the execution price is the seller's limit (`lib.rs:462`; `instructions/sharded_match_orders.rs:40`):

$$p^* = p_s \tag{3}$$

i.e. the full bid–ask spread accrues to the buyer, a deliberately conservative rule for a market in which buyers are predominantly consumers.

Off-chain matching (settled on-chain). The off-chain CDA engine is free to choose any execution price, but settlement enforces that the price lies within the limits **signed by both counterparties** in their Ed25519 payloads (`instructions/settle_offchain.rs:718–719`, `SlippageExceeded`):

$$p_s \le p^* \le p_b \tag{4}$$

so no participant can be settled at a price worse than the limit they signed, regardless of engine behaviour. The reference engine splits the spread at the midpoint, $p^* = \lfloor (p_b + p_s)/2 \rfloor$ (`scripts/simulate-market-clearing.ts:131`), which divides the surplus equally between the counterparties.

Uniform-price epoch clearing. Accumulated bids clear at a single price per 15-minute epoch. The oracle admits a clearing trigger only on an exact 900-second boundary strictly newer than the last cleared epoch (`programs/oracle/src/lib.rs:202–212`), and each executed match records its price as the zone's `last_clearing_price` (`programs/trading/src/lib.rs:494`; `settle_offchain.rs:946`).

Price transparency. The market maintains a 24-slot ring buffer of recent trade prices and recomputes a volume-weighted average price on each update (`programs/trading/src/lib.rs:820–863`):

$$\mathrm{VWAP} = \left\lfloor \frac{\sum_i p_i v_i}{\sum_i v_i} \right\rfloor \tag{5}$$

Network charges (market fee, wheeling, and loss, §11.3) are levied on top of the execution price and are bounded by the 20% cap of Equation 10, so the effective delivered price a buyer faces is p^* plus a capped, auditable surcharge.

11.3 Trade settlement

Gross value — rescale the 10^{15}-scaled product (9-dec energy × 6-dec price) back to 6-dec currency by the energy divisor $D_E = 10^9$ (`settle_offchain.rs:1145`):

$$V = \left\lfloor \frac{q \cdot p}{10^9} \right\rfloor \tag{6}$$

Market fee (`settle_offchain.rs:788`):

$$F_m = \left\lfloor \frac{V \cdot \varphi_m}{10^4} \right\rfloor \tag{7}$$

Wheeling — flat per-kWh, **not** ad-valorem; same energy rescale as Equation 6 (`settle_offchain.rs:1160`):

$$C_w = \left\lfloor \frac{q \cdot w}{10^9} \right\rfloor \tag{8}$$

Line loss (`settle_offchain.rs:1189`):

$$C_\ell = \left\lfloor \frac{V \cdot \ell}{10^4} \right\rfloor \tag{9}$$

Network charge and its cap invariant, $\Theta = 2000$ bps (20%); `ChargesExceedCap` if violated (`settle_offchain.rs:116`):

$$C_{\text{net}} = C_w + C_\ell, \quad C_{\text{net}} \le \left\lfloor \frac{V \cdot \Theta}{10^4} \right\rfloor \tag{10}$$

Seller net proceeds, with feasibility $F_m + C_{\text{net}} \le V$ enforced (`ChargesExceedValue`, `settle_offchain.rs:123`):

$$N_{\text{seller}} = V - F_m - C_{\text{net}} \tag{11}$$

11.4 Treasury peg (GRX ↔ THBG)

Swap GRX → THBG, divisor $D_G = 10^9$ atoms/whole-GRX (`treasury/src/lib.rs:58`):

$$G = \left\lfloor \frac{g_{\text{in}} \cdot r}{10^9} \right\rfloor, \quad F = \left\lfloor \frac{G \cdot \varphi_s}{10^4} \right\rfloor, \quad n = G - F \tag{12}$$

Peg invariants — reserve attestation freshness and full backing (`PegBreach`, `treasury/src/lib.rs:70`):

$$t_{\text{now}} - t_{\text{att}} \le \tau_{\text{ttl}}, \quad S_{\text{thbg}} + n \le R_{\text{att}} \tag{13}$$

Redeem THBG → GRX with collateral guards $a_{\text{in}} \le S_{\text{thbg}}$ (`SupplyUnderflow`) and $g_{\text{out}} \le V_{\text{swap}}$ (`InsufficientVault`) (`treasury/src/lib.rs:91`):

$$g_{\text{out}} = \left\lfloor \frac{a_{\text{in}} \cdot 10^9}{r} \right\rfloor \tag{14}$$

11.5 MasterChef staking (GRX yield)

Accumulator update on `fund_rewards`, precision $\Pi = 10^{12}$, requires $T > 0$ (`treasury/src/lib.rs:975`):

$$A \leftarrow A + \left\lfloor \frac{a \cdot 10^{12}}{T} \right\rfloor \tag{15}$$

Pending reward and debt baseline for a position of size s; Π absorbs the division loss (exactness unit-tested, `lib.rs:126`):

$$\rho = \max\left(0, \left\lfloor \frac{s \cdot A}{10^{12}} \right\rfloor - d\right), \quad d = \left\lfloor \frac{s \cdot A}{10^{12}} \right\rfloor \tag{16}$$

11.6 Oracle validation (15-min epoch)

Anomaly gate — integer cross-multiplication avoids float division, κ = `max_production_consumption_ratio` (`oracle/src/lib.rs:462`):

$$E_{\text{prod}} \cdot 100 \le \kappa \cdot E_{\text{cons}} \tag{17}$$

Reading quality score (`oracle/src/lib.rs:370`):

$$Q = \min\left(100, \left\lfloor \frac{100 \cdot n_{\text{valid}}}{n_{\text{valid}} + n_{\text{reject}}} \right\rfloor\right) \tag{18}$$

Range admission: $E_{\min} \le E \le E_{\max}$.

11.7 Registry sharding & slashing

Deterministic shard from the first byte of the authority key k (`registry/src/lib.rs:71`):

$$\sigma(k) = k[0] \bmod 16 \tag{19}$$

Validator-active predicate, $\beta_{\min} = 10^{13}$ atoms $= 10\{,\}000$ GRX (`registry/src/lib.rs:35`):

$$\text{active} \iff s_{\text{grx}} \ge \beta_{\min} \tag{20}$$

11.8 Parameters (data)

Parameter	Value	Meaning
D_E, D_G	10^9	energy / GRX atomic divisor
Π	10^{12}	staking accumulator precision
Θ	2000 bps	max network-charge cap (20%)
$\beta_{\min}$	10^{13} atoms	min validator stake (10k GRX)
shards	16	registry counter shards
epoch	900 s	oracle market-clearing window
GRID/GRX dec	9	1 kWh = 10^9 atoms
THBG dec	6	1 THB = 10^6 minor
REC dec	6	1 token = 1 MWh

11.9 Fleet-ingest benchmark: metric definitions and test data

The meter-scaling benchmark of §9.3 is defined formally as follows. A run at fleet size N submits one transaction per meter per epoch over $E = 2$ epochs. Let N_{ok} denote confirmed transactions, and attribute every non-confirmed transaction to exactly one class: N_{send} (rejected at submission), N_{val} (executed, failed an oracle guard), or N_{exp} (accepted, never confirmed within τ_{conf}). Conservation holds by construction:

$$N = N_{\text{ok}} + N_{\text{send}} + N_{\text{val}} + N_{\text{exp}} \tag{21}$$

Confirmed goodput over a burst starting at t_0 with last observed confirmation at t_{last} (the value reported as TPS throughout §9):

$$\text{TPS} = \frac{N_{\text{ok}}}{t_{\text{last}} - t_0} \tag{22}$$

Per-transaction latency is $L_i = t_i^{\text{conf}} - t_i^{\text{send}}$, quantised by the status-poll period τ_{poll} (reported: median, 95th percentile, maximum). The loss decomposition of Table 4 separates the validation-rejection rate from true delivery loss:

$$\nu = \frac{N_{\text{val}}}{N}, \quad \delta = \frac{N_{\text{send}} + N_{\text{exp}}}{N} \tag{23}$$

Because resubmission is idempotent (§9.3), k independent retry rounds reduce effective delivery loss to δ^{k+1}; the measured worst case $\delta = 0.0035$ (100,000 meters) yields $\delta^2 \approx 1.2 \times 10^{-5}$ after a single retry.

The synthetic workload assigns meter i the reading pair

$$E_{\text{prod}}(i) = 80 + (7i \bmod 420), \quad E_{\text{cons}}(i) = 40 + (3i \bmod 210) \tag{24}$$

which deliberately drives a fixed ≈0.47% of indices past the anomaly gate (Equation 17 with $\kappa = 1000$): those transactions fail with `AnomalousReading`, at a fee, in every epoch — a deterministic, in-band probe that the oracle's input validation continues to fire under full fleet load. The per-transaction compute figures (Table 4) are sampled from the tail of each epoch's confirmed set, because the rotating ledger prunes older transaction history under loads above roughly 10^5 transactions.

Harness parameterisation (data):

Parameter	Value	Meaning
N	80 … 10^5	fleet size (distinct meter PDAs)
E	2	epochs per run (init + steady-state)
epoch spacing	$\geq$ 61 s	on-chain `min_reading_interval` + 1 (on-chain clock)
W	3,000 tx	in-flight window (unconfirmed cap)
τ_{poll}	1.5 s	bulk `getSignatureStatuses` sweep period
τ_{conf}	90 s	confirmation deadline before a send is `expired`
blockhash refresh	10 s	cached recent-blockhash renewal

Parameter	Value	Meaning
send workers	64	concurrent submission tasks
CU sample	≤200 tx/epoch	tail-of-burst `getTransaction` sample
retry	none	loss figures are first-attempt delivery
commitment	`confirmed`	counting threshold for N_{ok}

Harness: `scripts/bench-meter-throughput.ts`; per-run artifacts under `test-results/meter-throughput-<N>m-<timestamp>.{json,md}` with the combined tables in `test-results/meter-scaling-summary.md`.